Let's see how this forward pass corresponds to the schematic back in figure 10.13. There are a few novelties used in this code that have not come up elsewhere in this book and that you may be unaware of. One is the use of the `LayerNorm` layer in PyTorch, which unsurprisingly stands for *layer normalization*.

`LayerNorm` is one form of neural network normalization; another popular one is called *batch normalization* (or just `BatchNorm`). The problem with unnormalized neural networks is that the magnitude of the inputs to each layer in the neural network can vary dramatically, and the range of values that the inputs can take can change from batch to batch. This increases the variability of the gradients during training and leads to instability, which can significantly slow training. Normalization seeks to keep all inputs at each major step of computation to within a relatively fixed, narrow range (i.e., with some constant mean and variance). This keeps gradients more stable and can make training much faster.

As we have been discussing, self-attention (and the broader class of relational or graph) models are capable of feats that ordinary feedforward models struggle with due to their inductive bias of data being relational. Unfortunately, because the model involves a softmax in the middle, this can make training unstable and difficult as the softmax restricts outputs to within a very narrow range that can become saturated if the input is too big or small. Thus it is critical to include normalization layers to reduce these problems, and in our experiments `LayerNorm` improves training performance substantially, as expected.

10.3.3. Tensor contractions and Einstein notation

The other novelty in this code is the use of the `torch.einsum` function. Einsum is short for *Einstein summation* (also called *Einstein notation*); it was introduced by Albert Einstein as a new notation for representing certain kinds of operations with tensors. While we could have written the same code without Einsum, it is much simpler with it, and we encourage its use when it offers improved code readability.